

Name

M#

Models

Review $y' = A y + b$

Chapter 4:

Driven Mass-Spring system

$$m y'' + c y' + k y = f_0 \sin(\omega t)$$

The ODE

$$m y'' + c y' + k y = f_0 \sin(\omega t)$$

is the standard mass-spring model. The mass and spring are defined by the positive constants mass m , linear damping coefficient c , and spring constant k . The sinusoidal external force is described by the magnitude f_0 and the angular frequency ω . The damping c is usually small and we are interested in the response of the system to a broad range of angular frequencies from slow $\omega \ll 1$ to fast $\omega \gg 1$. We are going to see that there is a single angular frequency ω_0 near which small forces can produce outsized motions.

Complementary Solution

Particular Solution

General Solution and Interpretation

$$y'' + \frac{c}{m} y' + \frac{k}{m} y = \frac{f_0}{m} \sin(\omega t)$$

The general solution for the physically interesting underdamped case $0 < \frac{c}{2m} < \omega_0 = \sqrt{\frac{k}{m}}$ is

$$y = e^{-\frac{c}{2m} t} (c_1 \cos(\omega_c t) + c_2 \sin(\omega_c t)) + a_1 \cos(\omega t) + a_2 \sin(\omega t)$$

where $\left\| \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \right\|^2 = \frac{f_0^2/m^2}{(\omega^2 - \omega_0^2)^2 + (c/m)^2 \omega^2}$. In other words, the response is an exponentially decaying transient with a steady state response that can be large if the damping is small and the driving frequency is near the natural frequency of the system.

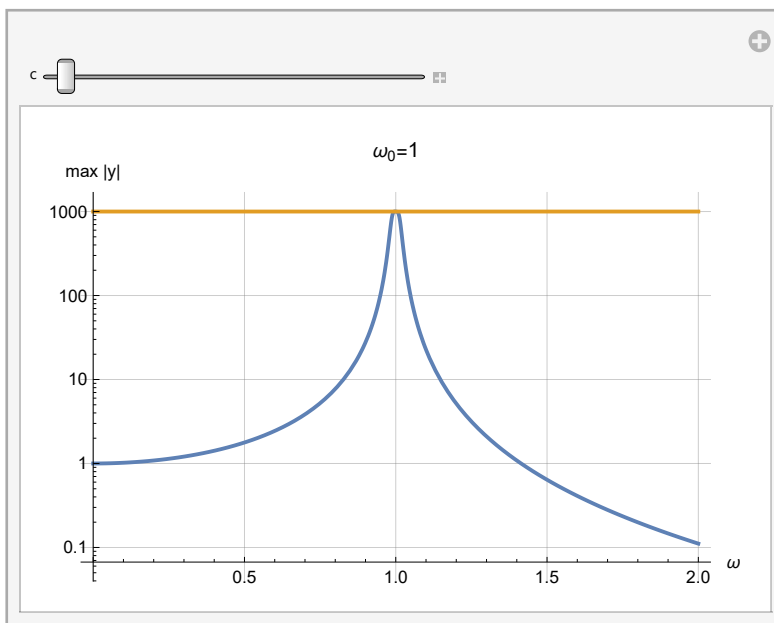
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In[16]:= m = 1;
ω₀ = 1;
Manipulate[
  LogPlot[ $\left\{ \sqrt{\frac{1}{(\omega^2 - \omega_0)^4 + \left(\frac{c}{m}\right)^2 \omega^2}}, \frac{1}{\left(\frac{c}{m}\right) \omega_0} \right\}$ , {ω, 0, 2 ω₀}, PlotRange → All,

  AxesLabel → {"ω", "max |y|"},
  PlotLabel → "ω₀=1", PlotPoints → 100, GridLines → Automatic],
  {{c, 0.001}, 0, 0.1}]

```

Out[18]=



Simulation with $c = 0$

$$m y'' + 0 y' + k y = f_0 \sin(\omega t)$$

When $c = 0$ the complementary solution

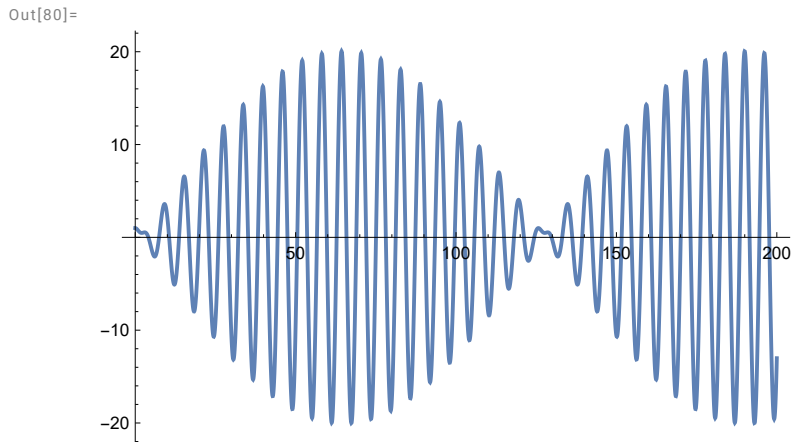
$$y_c = c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t)$$

is undamped and the ICs are no longer “transient”!

```

In[77]:= {m, c, k, f0, ω} = {1.0, 0, 1.0, 1.0, 1.05};
TMax = 200;
ySol = NDSolveValue[{
  m y''[t] + c y'[t] + k y[t] == f0 Sin[ω t],
  y[0] == 1,
  y'[0] == 0}, y, {t, 0, TMax}];
Plot[ySol[t], {t, 0, TMax}]

```



Here is a short audio demonstration of this “beats” phenomena of two almost equal frequencies.

<https://www.youtube.com/watch?v=V8W4Djz6jnY>

The “beats” come from the trig addition formulas

$$\sin \theta + \sin \varphi = 2 \sin \frac{1}{2}(\theta + \varphi) \cos \frac{1}{2}(\theta - \varphi),$$

$$\sin \theta - \sin \varphi = 2 \cos \frac{1}{2}(\theta + \varphi) \sin \frac{1}{2}(\theta - \varphi),$$

$$\cos \theta + \cos \varphi = 2 \cos \frac{1}{2}(\theta + \varphi) \cos \frac{1}{2}(\theta - \varphi),$$

$$\cos \theta - \cos \varphi = -2 \sin \frac{1}{2}(\theta + \varphi) \sin \frac{1}{2}(\theta - \varphi).$$

https://en.wikipedia.org/wiki/List_of_trigonometric_identities#Sum-to-product_identities